

Unit 14: Implicit Differentiation — Full Solutions

Warm-up

1. $x^2 + y^2 = 25$

Differentiate both sides with respect to x :

$$2x + 2y \cdot y' = 0$$

Solve for y' :

$$2y \cdot y' = -2x \quad \Rightarrow \quad y' = -\frac{x}{y}$$

2. $x^2 + 4y^2 = 16$

$$2x + 8y \cdot y' = 0$$

$$y' = -\frac{2x}{8y} = -\frac{x}{4y}$$

3. $y^2 = 4x$

$$2y \cdot y' = 4$$

$$y' = \frac{4}{2y} = \frac{2}{y}$$

4. $x^2 - y^2 = 9$

$$2x - 2y \cdot y' = 0$$

$$2y \cdot y' = 2x \quad \Rightarrow \quad y' = \frac{x}{y}$$

5. $xy = 6$

Use the product rule on the left side:

$$y + x \cdot y' = 0$$

$$y' = -\frac{y}{x}$$

6. $x + y^2 = 10$

$$1 + 2y \cdot y' = 0$$

$$y' = -\frac{1}{2y}$$

$$7. \ 3x^2 + 2y^2 = 12$$

$$6x + 4y \cdot y' = 0$$

$$y' = -\frac{6x}{4y} = -\frac{3x}{2y}$$

Standard

$$8. \ x^2y + xy^2 = 8$$

Differentiate each term using the product rule.

$$\frac{d}{dx}[x^2y] = 2xy + x^2y'$$

$$\frac{d}{dx}[xy^2] = y^2 + x \cdot 2y \cdot y' = y^2 + 2xy \cdot y'$$

The right side's derivative is 0 (it's a constant):

$$2xy + x^2y' + y^2 + 2xy \cdot y' = 0$$

Collect the y' terms:

$$y'(x^2 + 2xy) = -(2xy + y^2)$$

$$y' = \frac{-(2xy + y^2)}{x^2 + 2xy}$$

$$9. \ 2xy + y^2 + x^2 = x + y$$

Differentiate term by term.

$$\frac{d}{dx}[2xy] = 2(y + x \cdot y') = 2y + 2xy'$$

$$\frac{d}{dx}[y^2] = 2y \cdot y'$$

$$\frac{d}{dx}[x^2] = 2x$$

Right side: $\frac{d}{dx}[x+y] = 1+y'$

$$2y+2xy'+2yy'+2x=1+y'$$

Collect y' terms on the left, everything else on the right:

$$2xy'+2yy'-y'=1-2y-2x$$

$$y'(2x+2y-1)=1-2y-2x$$

$$y'=\frac{1-2y-2x}{2x+2y-1}$$

10. $xy^2+x^2y=6$

$$\frac{d}{dx}[xy^2]=y^2+2xy\cdot y'$$

$$\frac{d}{dx}[x^2y]=2xy+x^2y'$$

$$y^2+2xy\cdot y'+2xy+x^2y'=0$$

$$y'(2xy+x^2)=-(y^2+2xy)$$

$$y'=\frac{-(y^2+2xy)}{2xy+x^2}$$

11. $x^2y^3=x+y$

$$\frac{d}{dx}[x^2y^3]=2xy^3+x^2\cdot 3y^2\cdot y'=2xy^3+3x^2y^2y'$$

$$2xy^3+3x^2y^2y'=1+y'$$

Collect y' terms:

$$3x^2y^2y'-y'=1-2xy^3$$

$$y'(3x^2y^2-1)=1-2xy^3$$

$$y'=\frac{1-2xy^3}{3x^2y^2-1}$$

$$12. \ y^3 + x^2y = 2x$$

$$\frac{d}{dx}[y^3] = 3y^2y'$$

$$\frac{d}{dx}[x^2y] = 2xy + x^2y'$$

$$3y^2y' + 2xy + x^2y' = 2$$

$$y'(3y^2 + x^2) = 2 - 2xy$$

$$y' = \frac{2 - 2xy}{3y^2 + x^2}$$

$$13. \ xy + y^2 = 4$$

$$\frac{d}{dx}[xy] = y + xy'$$

$$\frac{d}{dx}[y^2] = 2yy'$$

$$y + xy' + 2yy' = 0$$

$$y'(x + 2y) = -y$$

$$y' = -\frac{y}{x + 2y}$$

$$14. \ x^3 + y^3 = 6xy$$

$$\frac{d}{dx}[x^3] = 3x^2$$

$$\frac{d}{dx}[y^3] = 3y^2y'$$

$$\frac{d}{dx}[6xy] = 6(y + xy') = 6y + 6xy'$$

$$3x^2 + 3y^2y' = 6y + 6xy'$$

Collect y' terms:

$$3y^2y' - 6xy' = 6y - 3x^2$$

$$y'(3y^2 - 6x) = 6y - 3x^2$$

Divide top and bottom by 3:

$$y' = \frac{2y - x^2}{y^2 - 2x}$$

Challenge

15. $y \sin\left(\frac{1}{x}\right) = 1 - xy$

Left side (product rule, with the second factor needing the chain rule):

Let $u = y$ ($u' = y'$) and $v = \sin\left(\frac{1}{x}\right)$. To find v' , use the chain rule with inner function $\frac{1}{x} = x^{-1}$ (derivative $-x^{-2} = -\frac{1}{x^2}$):

$$v' = \cos\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) = -\frac{\cos(1/x)}{x^2}$$

$$\frac{d}{dx} \left[y \sin\left(\frac{1}{x}\right) \right] = y' \sin\left(\frac{1}{x}\right) + y \cdot \left(-\frac{\cos(1/x)}{x^2}\right) = y' \sin\left(\frac{1}{x}\right) - \frac{y \cos(1/x)}{x^2}$$

Right side:

$$\frac{d}{dx} [1 - xy] = 0 - (y + xy') = -y - xy'$$

Set both sides equal:

$$y' \sin\left(\frac{1}{x}\right) - \frac{y \cos(1/x)}{x^2} = -y - xy'$$

Collect the y' terms on the left:

$$y' \sin\left(\frac{1}{x}\right) + xy' = -y + \frac{y \cos(1/x)}{x^2}$$

$$y' \left[\sin\left(\frac{1}{x}\right) + x \right] = y \left[\frac{\cos(1/x)}{x^2} - 1 \right]$$

$$y' = \frac{y \left[\frac{\cos(1/x)}{x^2} - 1 \right]}{\sin\left(\frac{1}{x}\right) + x}$$

16. $y^3 = \frac{x-1}{x+1}$

Left side: $\frac{d}{dx} [y^3] = 3y^2 y'$

Right side: use the quotient rule with $f = x - 1$ ($f' = 1$), $g = x + 1$ ($g' = 1$):

$$\frac{d}{dx} \left[\frac{x-1}{x+1} \right] = \frac{1 \cdot (x+1) - (x-1) \cdot 1}{(x+1)^2} = \frac{(x+1) - (x-1)}{(x+1)^2} = \frac{2}{(x+1)^2}$$

Set both sides equal:

$$3y^2 y' = \frac{2}{(x+1)^2}$$

$$y' = \frac{2}{3y^2(x+1)^2}$$

17. $x^2(x+y)^2 = x^2 + y^2$

Left side: product rule with $f = x^2$ ($f' = 2x$) and $g = (x+y)^2$. For g' , use the chain rule with inner function $x+y$ (whose derivative is $1+y'$, since y depends on x):

$$g' = 2(x+y)(1+y')$$

$$\frac{d}{dx} [x^2(x+y)^2] = 2x(x+y)^2 + x^2 \cdot 2(x+y)(1+y')$$

Expand the second piece: $2x^2(x+y)(1+y') = 2x^2(x+y) + 2x^2(x+y)y'$.

So the left side becomes:

$$2x(x+y)^2 + 2x^2(x+y) + 2x^2(x+y)y'$$

Right side:

$$\frac{d}{dx} [x^2 + y^2] = 2x + 2y \cdot y'$$

Setting both sides equal:

$$2x(x+y)^2 + 2x^2(x+y) + 2x^2(x+y)y' = 2x + 2y \cdot y'$$

Collect the y' terms:

$$2x^2(x+y)y' - 2yy' = 2x - 2x(x+y)^2 - 2x^2(x+y)$$

$$y' [2x^2(x+y) - 2y] = 2x - 2x(x+y)^2 - 2x^2(x+y)$$

Divide top and bottom by 2:

$$y' = \frac{x - x(x+y)^2 - x^2(x+y)}{x^2(x+y) - y}$$

18. $\sin(xy) = x + y$

Left side: chain rule with inner function $u = xy$ (whose derivative uses the product rule: $u' = y + xy'$):

$$\frac{d}{dx}[\sin(xy)] = \cos(xy) \cdot (y + xy')$$

Right side:

$$\frac{d}{dx}[x + y] = 1 + y'$$

Set equal:

$$\cos(xy) \cdot (y + xy') = 1 + y'$$

Expand the left side:

$$y \cos(xy) + xy' \cos(xy) = 1 + y'$$

Collect y' terms:

$$xy' \cos(xy) - y' = 1 - y \cos(xy)$$

$$y'[x \cos(xy) - 1] = 1 - y \cos(xy)$$

$$y' = \frac{1 - y \cos(xy)}{x \cos(xy) - 1}$$

19. $x^2y + y^2 = 3xy^2$

$$\frac{d}{dx}[x^2y] = 2xy + x^2y'$$

$$\frac{d}{dx}[y^2] = 2yy'$$

$$\frac{d}{dx}[3xy^2] = 3(y^2 + x \cdot 2y \cdot y') = 3y^2 + 6xyy'$$

$$2xy + x^2y' + 2yy' = 3y^2 + 6xyy'$$

Collect the y' terms:

$$x^2y' + 2yy' - 6xyy' = 3y^2 - 2xy$$

$$y'(x^2 + 2y - 6xy) = 3y^2 - 2xy$$

$$y' = \frac{3y^2 - 2xy}{x^2 + 2y - 6xy}$$

20. Tangent line to $x^2 + y^2 = 25$ at $(3, 4)$.

From Problem 1, we already know: $y' = -\frac{x}{y}$.

At $(3, 4)$:

$$y' = -\frac{3}{4}$$

Using point-slope form with point $(3, 4)$ and slope $-\frac{3}{4}$:

$$y - 4 = -\frac{3}{4}(x - 3)$$

$$y = -\frac{3}{4}x + \frac{9}{4} + 4 = -\frac{3}{4}x + \frac{9}{4} + \frac{16}{4} = -\frac{3}{4}x + \frac{25}{4}$$

Tangent line: $y = -\frac{3}{4}x + \frac{25}{4}$.

Applied

21. $4x + 9y^2 = 100$

$$4 + 18y \cdot y' = 0$$

$$y' = -\frac{4}{18y} = -\frac{2}{9y}$$

Answer: $\frac{dy}{dx} = -\frac{2}{9y}$.

22. $p^2 + q^2 = 200$, find $\frac{dp}{dq}$.

Differentiate with respect to q , treating p as a function of q :

$$2p \cdot \frac{dp}{dq} + 2q = 0$$

$$\frac{dp}{dq} = -\frac{2q}{2p} = -\frac{q}{p}$$

Answer: $\frac{dp}{dq} = -\frac{q}{p}$.

23. $x^2 + xy + y^2 = 19$

Differentiate:

$$2x + (y + xy') + 2yy' = 0$$

$$2x + y + xy' + 2yy' = 0$$

Collect y' terms:

$$y'(x + 2y) = -(2x + y)$$

$$y' = -\frac{2x + y}{x + 2y}$$

Check the point $(2, 3)$: $2^2 + 2(3) + 3^2 = 4 + 6 + 9 = 19$.

Evaluate at $(2, 3)$:

$$y' = -\frac{2(2) + 3}{2 + 2(3)} = -\frac{4 + 3}{2 + 6} = -\frac{7}{8}$$

Answer: $\frac{dy}{dx} = -\frac{2x + y}{x + 2y}$, and at $(2, 3)$, $\frac{dy}{dx} = -\frac{7}{8}$. This means the boundary curve of the machine part has a slope of $-\frac{7}{8}$ at that point.

24. $xy + x^2 = 12$

Differentiate:

$$y + xy' + 2x = 0$$

$$xy' = -y - 2x$$

$$y' = -\frac{y + 2x}{x}$$

Find y when $x = 2$: plug into the original equation, $2y + 4 = 12 \Rightarrow 2y = 8 \Rightarrow y = 4$. So the point is $(2, 4)$.

Evaluate at $(2, 4)$:

$$y' = -\frac{4 + 2(2)}{2} = -\frac{4 + 4}{2} = -\frac{8}{2} = -4$$

Answer: $\frac{dy}{dx} = -\frac{y + 2x}{x}$, and at $(2, 4)$, $\frac{dy}{dx} = -4$. This means the marginal rate of substitution at that point on the indifference curve is -4 .